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TESTING FOR "MONOPOLY" EQUILIBRIUM*

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This paper develops a very general test for "monopoly". Using standard comparative statics analysis, we derive testable restrictions on the firm's reduced form revenue equation which must be satisfied by any profit maximizing firm whose choices are not affected by either strategic interactions or the threat of entry. For such an unfettered monopolist, the sum of the factor price elasticities of the reduced form revenue equation must be nonpositive. The set of interesting alternative hypotheses is not empty. We develop simple models of oligopolistic, competitive and monopolistically competitive markets for which this test statistic may take on positive values.

I. INTRODUCTION

THE UNIFYING theme of this issue is empirical studies of firm and industry behavior. Unlike the other papers included here, we have no new empirical results to offer. However our analysis is a direct result of a puzzling empirical regularity we encountered over a decade ago. In attempting to find a theoretical explanation for that result, we derived a generally applicable testable implication of monopoly profit maximizing behavior. We think that both the test itself and the comparative statics approach used to derive it may be of interest to those, like ourselves, interested in investigating the performance of industrial markets using firm and industry level data. The tests are based on properties of reduced form revenue equations at the firm level, and the data requirements (revenues and factor prices) are relatively modest.

In the course of an investigation of the American daily newspaper industry, one of us (Rosse) noticed that the coefficients of newsprint and other factor prices were persistently positive in the reduced form revenue equations which he was estimating. Intuitively, this seemed very strange. The vast majority of newspapers are local monopolies, and the textbook theory of the unconstrained monopolist would suggest that an upward shift in its marginal cost curve would lead to a reduction of both equilibrium output and revenues. While the issue turns out to be somewhat more complex when dealing with multi-input, multi-output firms, we were clearly on the right track: the hypothesis of monopoly profit maximization does place testable restrictions

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on the estimated parameters of firms' reduced form revenue equations. Section II provides a general analysis of this issue.

However that left our theoretical work only partially complete. For example, even if one found that the parameter estimates of daily newspapers' reduced form revenue equations were not consistent with the hypothesis of monopoly profit maximization, it would then be necessary to investigate whether or not there exist economically interesting alternative hypotheses which were consistent with the data. This issue is relevant to hypothesis testing generally, for if *all* interesting economic models lead to the same testable restrictions, what is learned from rejection?

Section III develops simple models of equilibrium in single product monopolistically competitive, oligopolistic and perfectly competitive markets. Then, using standard comparative static techniques, we demonstrate that these models yield testable restrictions on firms' reduced form revenue equations which are different from those implied by the monopoly model. In all cases, the implications are in terms of the response of the equilibrium values of firm revenues to changes in factor prices.

While the monopoly model is quite general, the models of oligopoly, monopolistic and perfect competition that we analyze are as simple and as devoid of complications as possible. We present these models primarily to provide economically interesting alternatives to the monopoly hypothesis and to illustrate the generality of our methodological approach. In an actual empirical application it would be desirable to incorporate more specific institutional details and prior information in order to obtain as sharp a statistical test as possible. However, from a purely positive standpoint, it is probably always a useful starting point to determine which simple "as if" model best explains the behavior of the firms under study.

Finally, we should explain why "monopoly" appears in quotation marks in the title of this paper. This was done to emphasize one of the most important aspects of our analysis. For the refutable implications derived in section II apply to any firm whose *structural* revenue function contains as *endogenous* variables *only* those under the direct control of the firm. While this certainly captures the traditional definition of monopoly, it also applies to price-taking competitive firms, as long as the prices they face are truly exogenous, that is, as long as their equilibrium values are unaffected by changes in the other exogenous variable in the model. Thus an empirical refutation of our "monopoly" equilibrium hypothesis constitutes a rejection of the assumption that the revenues of the firms in question are independent of the decisions of rivals, actual or potential.

II. A TESTABLE IMPLICATION OF "MONOPOLY" EQUILIBRIUM

Let $\mathbf{y}$ be a vector of decision variables which affect the firm's revenues, so that $R = R(\mathbf{y}, \mathbf{z})$, where $\mathbf{z}$ is a vector of exogenous variables that shift the firm's revenue function. In addition, we assume that the firm's costs also depend, directly or indirectly, on $\mathbf{y}$, so that $C = C(\mathbf{y}, \mathbf{w}, \mathbf{t})$. Here, $\mathbf{w}$ is a vector of m factor prices that are exogenous to the firm and $\mathbf{t}$ is a vector of exogenous variables that shift the firm's cost function. The vectors $\mathbf{t}$ and $\mathbf{z}$ may or may not have components in common. It is natural to think of $\mathbf{y}$ as representing a vector of output levels. However that interpretation is overly restrictive. The components of $\mathbf{y}$ could include prices, advertising expenditures or quality levels; i.e., any decision variables which *structurally* enter the monopolist's revenue function.[1]

Thus firm profits can be written as $\pi = R - C = \pi(\mathbf{y}, \mathbf{z}, \mathbf{w}, \mathbf{t})$. Let $\mathbf{y}^0 = \operatorname{argmax}_{\mathbf{y}}\{\pi(\mathbf{y}, \mathbf{z}, \mathbf{w}, \mathbf{t})\}$ and $\mathbf{y}^1 = \operatorname{argmax}_{\mathbf{y}}\{\pi(\mathbf{y}, \mathbf{z}, (1+h)\mathbf{w}, \mathbf{t})\}$, with the scalar $h \geqslant 0$. Also, let $R^0 = R(\mathbf{y}^0, \mathbf{z}) \equiv R^*(\mathbf{z}, \mathbf{w}, \mathbf{t})$ and $R^1 = R(\mathbf{y}^1, \mathbf{z}) \equiv R^*(\mathbf{z}, (1+h)\mathbf{w}, \mathbf{t})$, where R^* is the firm's reduced form revenue function. Then, by definition,

$$(1) \qquad R^1 - C(\mathbf{y}^1, (1+h)\mathbf{w}, \mathbf{t}) \geqslant R^0 - C(\mathbf{y}^0, (1+h)\mathbf{w}, \mathbf{t})$$

Because C is linearly homogeneous in $\mathbf{w}$, this can be written as

$$(2) \qquad R^1 - (1+h)C(\mathbf{y}^1, \mathbf{w}, \mathbf{t}) \geqslant R^0 - (1+h)C(\mathbf{y}^0, \mathbf{w}, \mathbf{t})$$

Similarly, it must also be the case that

$$(3) \qquad R^0 - C(\mathbf{y}^0, \mathbf{w}, \mathbf{t}) \geqslant R^1 - C(\mathbf{y}^1, \mathbf{w}, \mathbf{t})$$

Multiplying both sides of (3) by $1+h$ and adding the result to (2) yields

$$(4) \qquad -h(R^1 - R^0) \geqslant 0$$

Dividing both sides of (4) by $-h^2$, we obtain

$$(5) \qquad (R^1 - R^0)/h = [R^*(\mathbf{z}, (1+h)\mathbf{w}, \mathbf{t}) - R^*(\mathbf{z}, \mathbf{w}, \mathbf{t})]/h \leqslant 0$$

This nonparametric[2] version of our result simply states that a proportional cost increase always results in a decrease in the firm's revenues. Assuming that the reduced form revenue function is differentiable, taking the limit of (5) as $h \to 0$ and then dividing the result by R^* yields

$$(6) \qquad \psi^* \equiv \sum w_i(\partial R^*/\partial w_i)/R^* \leqslant 0$$

Thus we have established

Theorem 1: The sum of the factor price elasticities of a monopolist's reduced form revenue equation must be nonpositive.

[1] Care must be taken when interpreting the cost function when some or all of the components of $\mathbf{y}$ are prices. For example, if $\mathbf{y}$ is a price vector and $\mathbf{Q} = \mathbf{Q}(\mathbf{y}, \mathbf{z})$ is the associated system of market demand equations, then $C(\mathbf{y}, \mathbf{w}, \mathbf{t}) = \tilde{C}(\mathbf{Q}(\mathbf{y}, \mathbf{z}), \mathbf{w}, \boldsymbol{\tau})$, where $\tilde{C}$ is the conventional multi-product cost function, $\boldsymbol{\tau}$ is a vector of exogenous technological parameters (that do not enter the demand functions) and $\mathbf{t} = (\mathbf{z}, \boldsymbol{\tau})$. It is important to note in such cases that C (and its associated gradient) "inherit" linear homogeneity in $\mathbf{w}$ from the structural cost function $\tilde{C}$.

[2] We are indebted to Hal Varian for suggesting this line of proof. See Ashenfelter and Sullivan [1987] for a nonparametric analysis of the implications of oligopoly behavioral models.

Intuitively, it is easy to trace the source of the power of this result. For ψ can, alternatively, be defined as the elasticity of reduced form revenues $R^*(\mathbf{z}, k, \mathbf{w}, \mathbf{t})$ with respect to the scalar $k = 1+h$, evaluated at $k = 1$. In words, this asks the question: What will be the percentage change in equilibrium revenues resulting from a 1% change in all factor prices? We know from duality theory that a 1% increase in all factor prices leads to precisely a 1% upward shift in *all* of the firm's cost curves: average, total, and marginal. Thus we can rephrase the above question to read: What will happen to the monopolist's revenues if its costs increase by 1%? In effect, estimating ψ provides a way to empirically shift the firm's cost curves even though cost data are unavailable!

An illustration of the practical application of this analysis can be obtained from the analysis of a simple single product monopoly model. Suppose that the monopolist faces a demand curve of constant price elasticity $e > 1$, so that $R(y, z) = \gamma z^{\alpha} y^{(e-1)/e}$. For simplicity, assume also that the monopolist employs a constant returns to scale Cobb–Douglas technology, so that its cost function can be written as

$$(7) \qquad [\ln C(y, \mathbf{w}, t)] = [\ln y] + \beta[\ln t] + \sum a_i[\ln w_i]; \qquad a_i > 0 \text{ and } \sum a_i = 1$$

Then it is straightforward to show that the (nonstochastic version of the) firm's reduced form revenue equation is given by

$$(8) \qquad [\ln R^*(z, t, \mathbf{w})] = \gamma_0 + e\alpha[\ln z] - (e-1)\beta[\ln t] - (e-1)\sum a_i[\ln w_i]$$

where the intercept $\gamma_0 = e[\ln\gamma] - (1-e)[\ln(e-1)/e]$. Thus $\psi^* = -(e-1)\sum a_i = 1-e$, which must be negative if the marginal cost curve is to cut the marginal revenue curve from below as required by the second order conditions for monopoly profit maximization. Note that, because of the simple structure of the example, in this case estimation of the reduced form revenue equation not only allows one to test the hypothesis of monopoly profit maximization, it also provides the analyst with estimates of all of the structural parameters of interest when the hypothesis is maintained. In particular, this example indicates that the *magnitude*, as well as the sign, of ψ^* may be of interest. Here, because it provides an estimate of the price elasticity of demand, ψ^* also yields an estimate of the Lerner index of monopoly power $L = (e-1)/e = \psi^*/(\psi^* - 1)$.

Theorem 1 is a very general result, requiring little beyond the maximization hypothesis itself. For example, it does not require that the observations be generated by long-run equilibrium behavior. Components of $\mathbf{z}$ and $\mathbf{t}$ might be decision variables of the firm that it is reasonable to assume are fixed, or predetermined, over the relevant time frame. However, it is important to recognize a limitation of the analysis stemming from the very generality of our result: ψ^* must be nonpositive for all "monopolies", even those facing a perfectly elastic market demand curve!

To see this intuitively, note that nothing in the analysis prevents $\mathbf{z}$ from being an exogenously given world price vector, so that $R = \mathbf{z} \cdot \mathbf{y}$. To take a specific single product example, let $R(y, z) = zy$ and assume that the firm operates a decreasing returns Cobb–Douglas technology, so that its cost function can be written as

$$(9) \qquad [\ln C(y, \mathbf{w}, t)] = \theta + \delta[\ln y] + \beta[\ln t] + \sum a_i[\ln w_i]: \quad \delta > 1; \theta, a_i > 0 \text{ and } \sum a_i = 1$$

Then it is straightforward to show that the (nonstochastic version of the) firm's reduced form revenue equation is given by

$$(10) \qquad [\ln R^*(z, t, \mathbf{w})] = \{\delta[\ln z] - \theta_0 - \beta[\ln t] - \sum a_i[\ln w_i]\}/(\delta - 1)$$

where $\theta_0 = \theta + [\ln \delta]$. Thus $\psi^* = \sum a_i/(\delta - 1) = 1/(\delta - 1) < 0$. In this case, while the reduced form revenue equation again yields estimates of all the structural parameters of interest, the magnitude of ψ^* is totally unrelated to the degree of monopoly power, which is zero by hypothesis.

The above examples should convince the reader that, while both of the firms in question were monopolists in the strict sense of the term, what they have in common is not the degree of monopoly power that they enjoy. Rather, the common element in both cases is that they operate in isolation: their structural revenue functions do not depend on any other optimizing agents' decision variables. Thus a rejection of the hypothesis that $\psi \leqslant 0$ must mean that the revenue functions of the observed firms are influenced by the actions of others. This intuition is made precise in the models of the next section.

III. ALTERNATIVE MODELS AND THEIR TESTABLE IMPLICATIONS

The analysis of the preceding section has established that an empirical investigation yielding an estimate of ψ significantly greater than 0 constitutes grounds for rejecting the hypothesis that the data were generated by firms acting as independent "monopolists". The natural next question, then, is whether or not there exist *any* models of firm and industry equilibrium consistent with estimated values of $\psi > 0$. If not, the above hypothesis test is of little practical use. Fortunately, the answer is in the affirmative. In this section we show that three commonly employed simple models of industry equilibrium are consistent with values of $\psi > 0$. As one would expect, all three models have the property that the revenue function facing the individual firm depends either implicitly or explicitly upon the decisions of its actual or potential rivals. Finally, it is important to note that, unlike the monopoly model, the results for the models of perfect and monopolistic competition depend quite crucially on the assumption that the firms in question are observed in long-run equilibrium.

III(i). *Monopolistic competition*

Recall for a moment the practical problem which caused us to begin our theoretical analyses. The daily newspapers whose reduced form revenue equations did not conform to the predictions of the monopoly equilibrium hypothesis were, for the most part, the only newspapers in their market areas. Thus, in order to model their behavior interactively, it was natural to posit that they were in competition with other purveyors of *differentiated* information services such as magazines and radio and television stations. The theory of monopolistic competition is therefore the source of our first alternative hypothesis.

Chamberlin [1962] adamantly maintained that firms selling differentiated products were qualitatively indistinguishable from classical monopolists:

> The more substitutes controlled by any one seller, the higher he can put his price. But that is another matter. As long as the substitutes are to any degree imperfect, he still has a monopoly of his own product and control over its price within the limits imposed upon any monopolist—those of the demand. (page 67)

Thus it was his position that one could *not* distinguish between a monopolistic competitor and a pure monopolist from observation of the firm's decision variables. However, at the same time, Chamberlin insisted that *market* equilibria in the two models were vastly different:

> The theory of monopoly, although the opening wedge, is very soon discovered to be inadequate.... Within any group of closely related products (such as that ordinarily included in one imperfectly competitive market) the demand and cost conditions (and hence the price) of any one are defined only if the demand and cost conditions with respect to the others are taken as given. Partial solutions of this sort, yielded by the theory of monopoly, contribute nothing towards a solution of the whole problem, for each rests upon assumptions with respect to the others. (pages 68 and 69)

In sum, he writes:

> Monopolistic competition is evidently a different thing from either *pure* monopoly or *pure* competition. As for monopoly, *as ordinarily conceived and defined*, monopolistic competition embraces it and takes it as a starting point. (page 68)
>
> Monopolistic competition, then, concerns itself not only with the problem of an *individual* equilibrium (the theory of ordinary monopoly), but also with that of a *group* equilibrium (the adjustment of economic forces within a group of competing monopolists, ordinarily regarded as a group of competitors. (page 69)

A moment's reflection should reveal that the comparative statics approach is ideally, and perhaps uniquely, suited to the task of developing testable hypotheses capable of distinguishing between these two theories. For only by analyzing the market equilibrium as a whole can we expect to find any differences. Each firm, viewed in isolation, would, in both theories, behave exactly as a monopoly would, and its actions would satisfy all the conditions of monopoly profit maximization. Nevertheless, it may be hoped that by examining the effects of changes in exogenous variables the "hidden" forces of Chamberlin's "group equilibrium" condition may come into focus.

In the past half century a good deal of the criticism of Chamberlinian theory has focused upon its inability to yield empirically refutable hypotheses. (G. C. Archibald [1961] provides perhaps the most scathing critique.) The conventional wisdom is thus that Chamberlin's model, *in the absence of any additional assumptions*, has no predictive power. However, by adding some additional structure to the model that is both plausible and present in most theoretical treatments, we are able to derive testable restrictions on the reduced form revenue equation of the monopolistically competitive firm.

What follows is an analysis of the comparative statics properties of the "Chamberlinian tangency". S. P. Das [1981] has studied this problem and its implications for the long-run behavior of the monopolistic competitors' output supply and input demand functions and contrasted them with the results of Silberberg [1974] and others on the behavior of the perfectly competitive firm in long-run equilibrium. However, as in the monopoly case, our analysis of the firm's reduced form revenue equation does not require detailed observations on output levels or prices.

In the familiar large group Chamberlinian equilibrium, the firm sets its price so that it operates at an output where perceived marginal revenue equals marginal cost. The demand curve facing the individual firm depends upon the prices (quantities) of the substitute products in the market. For expositional convenience, Chamberlin began the practice, much beloved by modern analysts, of imposing symmetry so that, while the products are differentiated, the representative firm can be studied diagrammatically. The entry or exit of additional products in response to profits or losses will cause the perceived demand curve of the representative firm to shift in or out until, in long-run equilibrium, zero economic profits are achieved with said demand curve tangent to the average cost curve. Under symmetry, these conditions allow us to define two equations that determine the equilibrium number of firms and their level of output.

As in the monopoly case, the exogenous variables whose impact we study are factor prices. Intuitively, the immediate effect of an input price increase is to shift upward the firm's average and marginal cost curves and reduce its output. However, this would result in losses for the representative firm, inducing exit by some firms, which would, in turn, result in an upward shift in the demand curve facing the representative firm until the tangency equilibrium was re-established.

To capture this process analytically and derive its testable implications, we first define the representative firm's perceived inverse demand function $P(y, n, \mathbf{z})$ that, under symmetry, relates price P to firm output y, the number of rivals n, and a vector of exogenous variables $\mathbf{z}$. We assume, as usual, that $\partial P/\partial y \equiv P_y < 0$ and that $\partial P/\partial n \equiv P_n < 0$. In addition, we require the eminently plausible:

Assumption 1: The elasticity of perceived demand facing the individual firm, $e(y, n, \mathbf{z}) \equiv -P/[y\partial P/\partial y]$, is a nondecreasing function of the number of (symmetric) rivals. That is, $\partial e/\partial n \geqslant 0$.

That the elasticity of demand facing the firm increases with the number (and in a symmetric model, perforce) the closeness of the substitutes with which it competes seems almost a truism. However, it involves some assumptions about the structure of the product space that are difficult to test directly. Here, we accept it on faith, noting only that the most commonly used theoretical models of monopolistic competition satisfy this condition.[3]

We now proceed to an analysis of the comparative statics properties of Chamberlinian equilibrium. Letting $R(y, n, \mathbf{z}) = yP(y, n, \mathbf{z})$, this is characterized by the two equation system

(11) $$R_y - C_y = 0$$

(12) $$R(y, \hat{n}, \mathbf{z}) - C(\hat{y}, \mathbf{w}, \mathbf{t}) = 0$$

Equations (11) and (12) implicitly define the long-run equilibrium values of $\hat{y}$ and $\hat{n}$ as functions of the exogenous variables $\mathbf{z}$, $\mathbf{w}$, and $\mathbf{t}$. Totally differentiating (11) and (12) with respect to w_i and solving using Cramer's Rule yields:

(13) $$\partial\hat{y}/\partial w_i = \{R_n(\partial\tilde{x}_i/\partial y) - R_{yn}\tilde{x}_i\}/\hat{D}$$

where the $\tilde{x}_i(y, \mathbf{w}, \mathbf{t})$ are conditional factor demand functions. We know that $\hat{D} \equiv (R_{yy} - C_{yy})R_n > 0$ from the second order conditions associated with (11). Since $\hat{R}(\mathbf{z}, \mathbf{w}, \mathbf{t}) = R(\hat{y}, \hat{n}, \mathbf{z})$, using (12) and the chain rule, we obtain

(14) $$\partial\hat{R}/\partial w_i = C_y(\partial\hat{y}/\partial w_i) + \tilde{x}_i$$

Multiplying by $(w_i/\hat{R})$ and summing (14) over all inputs yields

(15) $$\hat{\psi} = \sum w_i(\partial\hat{R}/\partial w_i)/\hat{R} = C/\hat{R} + (C_y/\hat{R})\sum w_i(\partial\hat{y}/\partial w_i)$$

Making use of (12) and (13), we have

(16) $$\hat{\psi} = 1 + C_y\{R_n \sum w_i(\partial\tilde{x}_i/\partial y) - R_{yn}\sum w_i\tilde{x}_i\}/\hat{R}\hat{D}$$

But from the definition of the cost minimizing input vector $\tilde{\mathbf{x}}$, this becomes

(17) $$\hat{\psi} = 1 + C_y\{R_n C_y - R_{yn}C\}/\hat{R}\hat{D}$$

[3] For example, the elasticity of the individual firm's perceived demand curve in the model of Dixit and Stiglitz [1977] is independent of the number of rivals, so that $\partial e/\partial n = 0$. This is also an implicit feature of the analysis of Spence [1976]. Those authors derive the firm's perceived demand curve under the assumption that the firm believes that total *industry* output will be unaffected by a change in its own output level. In the more general analysis of Koenker and Perry [1981], it can be shown that $\partial e/\partial n$ is positive when the firms hold less competitive (e.g. Cournot–Nash) conjectures about rivals' responses.

Substituting in the profit maximizing condition (11), the zero profit condition (12) and rearranging terms, we have

$$(18) \qquad \hat{\psi} = 1 + R_y[R_n R_y - R R_{yn}]/\hat{R}\hat{D}$$

Rewriting the bracketed term in (18) in terms of the inverse demand function and cancelling terms yields

$$(19) \qquad \hat{\psi} = 1 + R_y[y^2(P_n P_y - P P_{yn})]/\hat{R}\hat{D}$$

Since $\partial e/\partial n = (PP_{yn} - P_y P_n)/[y(P_y)^2]$, the bracketed term in (19) is non-positive by Assumption 1, and $\hat{\psi} \leqslant 1$. Thus we have established

Proposition 1: In symmetric Chamberlinian equilibrium, the sum of the elasticities of firm's reduced form revenues with respect to factor prices is less than or equal to unity.

The range of permissible values for $\hat{\psi}$ includes that of ψ^*(i.e., the negative real line) *plus* the unit interval. Thus the analyst can, in principle, observe data that are consistent with the hypothesis of monopolistic competition but *not* with that of profit maximizing monopoly.

III(ii). *Long-run competitive equilibrium*

A natural benchmark case to examine is that of perfect competition. However our analysis in section II makes it clear that price-taking behavior will lead to $\psi \leqslant 0$ unless some firm interactions are introduced into the model. Intuitively, the way to proceed is to formulate a model in which competitive entry makes the output price facing the firm an endogenous variable.

Over the last two decades there has arisen a literature on the theory of the competitive firm when observed in long-run equilibrium. The most elegant example of this literature is Silberberg [1974]. Briefly, it was found that the traditional Samuelsonian comparative statics of the firm must be modified when applied to long-run equilibrium behavior. This comes about for essentially the same reason as in the Chamberlinian case: changes in factor prices will, in the long-run, result in entry or exit and changes in the equilibrium output price, which will, in turn, affect the firm's input demand and output supply decisions. Here, we develop the implications of this long-run theory of the competitive firm in order to examine the properties of the associated reduced form revenue function.

The equilibrium price p^C and firm output level y^C in a perfectly competitive industry with free entry and a freely available technology is defined implicitly by the two equation system

$$(20) \qquad p^C - C_y(y^C, \mathbf{w}, \mathbf{t}) = 0$$

$$(21) \qquad p^C y^C - C(y^C, \mathbf{w}, \mathbf{t}) = 0$$

(A third equation, determining the equilibrium number of firms and aggregate industry output, is unnecessary for our present purposes.) Totally differentiating (20) and (21) with respect to w_i and solving via Cramer's Rule yields

(22) $$\partial y^C/\partial w_i = [\tilde{x}_i - y^C(\partial \tilde{x}_i/\partial y)]/y^C C_{yy}$$

Since $R^C(\mathbf{w}, \mathbf{t}) \equiv p^C(\mathbf{w}, \mathbf{t})\, y^C(\mathbf{w}, \mathbf{t})$, we obtain, using (21),

(23) $$\partial R^C/\partial w_i = C_y(\partial y^C/\partial w_i) + \tilde{x}_i$$

Multiplying (23) through by w_i and summing over all factors yields

(24) $$\sum w_i(\partial R^C/\partial w_i) = (C_y/yC_{yy})\{\sum w_i\tilde{x}_i - y\sum w_i(\partial \tilde{x}_i/\partial y)\} + \sum w_i\tilde{x}_i$$

Using the definitions of the total and marginal cost functions, this becomes

(25) $$\sum w_i(\partial R^C/\partial w_i) = (C_y/yC_{yy})\{C - yC_y\} + C$$

Substituting in (20) and (21) and dividing by R^C, we have

(26) $$\psi^C = \sum (w_i/R^C)(\partial R^C/\partial w_i) = 1$$

allowing us to state

Proposition 2: For firms observed in long-run competitive equilibrium, the *sum* of the elasticities of reduced form revenues with respect to factor prices equals unity.

The intuition behind this result should by now be clear, since ψ^C measures the impact of a proportional increase in all factor prices. Suppose all factor prices rise by 1%. Because average cost is linearly homogeneous in $\mathbf{w}$, such an increase will shift that curve upward by 1% for all output levels, leaving its minimum point unchanged. Since, in long-run competitive equilibrium firms always operate at said minimum , this means that y^C is unchanged. However, in equilibrium, p^C is equal to the level of AC at its minimum point, and this has increased by 1%. Thus equilibrium revenues also go up by 1%, the amount of the increase in factor prices. But this is precisely the condition that $\psi^C = 1$.

The astute reader will have noticed that there is another testable implication of this simple competitive model: reduced form revenues are not affected by shifts in the market demand curve; $\partial R^C/\partial z = 0$. This is due to the fact that we have been working with the simple textbook model of the long-run perfectly competitive industry which gives rise to a horizontal industry supply curve. In that model, equilibrium market price and output per firm are completely determined by technological variables and factor prices, so it should come as no surprise that equilibrium firm revenues are also unaffected by market demand conditions. However a simple extension of the model can handle the case of the competitive industry with a rising long-run supply curve.

Following standard textbook practice, suppose that the price of input 1 rises as the industry expands. That is, w_1 is exogenous to the firm but endogenous to the industry. Then, in addition to equations (16) and (17) we need to add conditions equating supply and demand in the output market ($Q(p^C, \mathbf{z}) = n^C y^C$) and in the market for input 1 (for example, $\hat{w}_1 = S[n^C \tilde{x}_1(y^C, \mathbf{w}, \mathbf{t}), \mathbf{s}]$). This would yield a system of four equations to determine the four endogenous variables p^C, y^C, n^C, and $\hat{w}_1$, where n^C is the long-run equilibrium number of firms in the industry. The firm's revenue equation, no longer a true reduced form, could be estimated as above, with the exogenous variables $\mathbf{z}$ and $\mathbf{s}$ available to construct an instrument for the endogenous variable $\hat{w}_1$.

Let us pause to review what has been accomplished thus far. We have analyzed three models of market equilibrium that yield distinctly different refutable implications with respect to the test statistic ψ. The "monopoly" equilibrium hypothesis requires that ψ be nonpositive, (symmetric) Chamberlinian equilibrium requires a $\psi \leqslant 1$, and in (symmetric) long-run competitive equilibrium it must be the case that $\psi = 1$. Thus, in any empirical application, rejecting the hypothesis of $\psi \leqslant 0$ rules out the "monopoly" model. Rejecting the hypothesis of $\psi \leqslant 1$ implies the rejection of all three models, while rejection of *both* the (hypothesis of $\psi \leqslant 0$ *and* the hypothesis of $\psi = 1$ (but not $\psi \leqslant 1$) means that, of the models considered so far, only the Chamberlinian model could be consistent with the data.[4]

III(iii). *Conjectural variation oligopoly*

The first two models in this section introduce interdependence (i.e., endogenous variables beyond the control of the firm) into firms' structural revenue functions via the hypothesis that free entry and exit result in zero profits in equilibrium. Next, we demonstrate that a model which exhibits strategic interactions among a *fixed* number of rivals may also be consistent with positive values of ψ.

Consider a simple, symmetric n firm homogeneous product oligopoly

[4] In the preliminary empirical analyses reported in Rosse and Panzar [1977], we were able to reject the "monopoly" hypothesis with t-levels ranging from 1.81 to 4.50, depending on the specification of the functional form of the reduced form revenue equation. For all but one specification, the hypothesis that $\psi = 1$ was rejected with t-levels greater than 2.5. The Chamberlinian hypothesis could not be rejected with any degree of confidence for any specification. (See tables IV and V.) Unfortunately these tests must be viewed as suggestive rather than conclusive. Since the firm data were reported anonymously, demographic data for the firms' local markets, such as population, *per capita* income, etc., were unavailable. Missing such important components of the vector $\mathbf{z}$, the reduced form revenue equation had no explanatory power. The impact of such missing variables were adequately reflected by the variation in the firms' circulation levels, and including circulation as an explanatory variable allowed us to obtain the estimates referred to above. However, for those specifications to yield valid results, circulation must be exogenously determined. Since the underlying structural model could not be identified, it was impossible to test this key assumption.

model. Each firm chooses its output level y_k to maximize its profits $\pi_k = P(Y, \mathbf{z})y_k - C(y_k, \mathbf{w}, \mathbf{t})$, where P is the market inverse demand curve and $Y = \sum y_k$ is total industry output. Then the first-order condition for profit maximization for the k^{th} *producing* firm requires

$$(27) \qquad \partial\pi_k/\partial y_k = \lambda y_k P_Y + P - C_y = 0$$

where the conjectural variation $\lambda = dY/dy_k$ measures the change in industry output which the k^{th} firm perceives will result from a marginal increase in y_k. (We assume that the second-order condition associated with (27) is also satisfied.) Typically, λ takes on values between 0 and n. If $\lambda = 0$, inspection of (27) reveals that the market behaves competitively. If $\lambda = n$, the collusive outcome is achieved. See Seade [1980] for a discussion of this class of oligopoly models.

Restricting attention to symmetric equilibria allows us to substitute the conditions $y_k = y^0$ and $Y^0 = ny^0$ into (27), obtaining

$$(28) \qquad \lambda y^0 P_Y(ny^0, \mathbf{z}) + P(ny^0, \mathbf{z}) - C_y(y^0, \mathbf{w}, \mathbf{t}) = 0$$

Equation (28) defines oligopolistic firm output y^0 as an implicit function of the variables λ, $\mathbf{z}$, $\mathbf{w}$, and $\mathbf{t}$. Then, *assuming that the degree of oligopolistic coordination, as measured by λ, is unaffected by changes in factor prices*, totally differentiating (28) with respect to w_i yields

$$(29) \qquad \partial y^0/\partial w_i = (\partial^2 C/\partial y \partial w_i)/D^0$$

where $D^0 = [ny^0\lambda P_{YY} + (n+\lambda)P_Y - C_{yy}]$. Seade [1980] has shown that $D^0 < 0$ is required for the stability of the symmetric equilibrium under study. Premultiplying both sides of (29) by w_i and summing over all i yields

$$(30) \qquad \sum w_i(\partial y^0/\partial w_i) = \sum w_i(\partial^2 C/\partial y \partial w_i)/D^0 = C_y/D^0 < 0$$

where the last equality, again, follows from the fact that marginal cost is linearly homogeneous in factor prices. Dividing by y^0 establishes

Proposition 3: For firms in a stable, symmetric, homogeneous product, conjectural variation oligopoly equilibrium, the *sum* of the factor price elasticities of the reduced form firm output equation, $y^0(\mathbf{z}, \mathbf{w}, \mathbf{t})$, is negative.

However the effects of factor prices on reduced form revenues will, in general, be indeterminate. Since $R^0(\mathbf{z}, \mathbf{w}, \mathbf{t}) = y^0 P(ny^0, \mathbf{z})$, applying the chain rule to (30) and dividing by R^0 yields

$$(31) \qquad \psi^0 = \sum w_i(\partial R^0/\partial w_i)/R^0 = (y^0 n P_Y + P)\sum w_i(\partial y^0/\partial w_i)/R^0 = R_Y C_y/D^0 R^0$$

where $R_Y = Y^0 P_Y + P$ is the equilibrium value of the *industry* marginal revenue curve. Now for an oligopoly which behaves sufficiently like a collusive monopoly (that is, $\lambda \approx n$), the first-order condition (31) will ensure that R_Y is positive. But, in general, there is nothing in this class of oligopoly models which restricts the industry's aggregate output to the elastic portion of the market demand curve.[5]

IV. CONCLUDING REMARKS

In this paper we have presented a methodology for uncovering some of the testable implications contained in models of firm and industry equilibrium. However it should be clear that we have not begun to exhaust the potential applications of positive, comparative statics approaches to industrial economics. We have assumed that the only firm specific data available are revenues and factor prices. It seems likely that much sharper tests would be possible if, as is sometimes the case, the analyst also has available data on equilibrium prices and quantities for the firm and/or industry.[6] The models examined here were selected for analysis because of their familiarity and tractability. However the cost minimization hypothesis and duality theory, the key ingredients in our comparative statics approach, can be employed in much the same way to virtually any system of well-defined equilibrium equations. The usefulness of focusing on the sum of the factor price elasticities of reduced form equations is that that statistic captures the effect of a proportional shift in the average, total, or marginal cost curve even when cost data themselves are unavailable. Thus the technique employed in this paper should prove useful whenever factor prices are observable but it is not possible to estimate cost functions.

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[5] This is not to say that there are no testable restrictions on the comparative statics properties of the model for various values of λ and n. See Sullivan [1985] for an analysis which successfully places bounds on the Cournot equivalent number of firms, n/λ.

[6] See Bresnahan [1982] and Lau [1982] for discussions of the conditions required to identify the conjectural variation parameter λ when observations on industry prices and quantities are available.

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